Arista Access Node WFD-ACC-44 Optical Transport Failure Resolution SOP

Node: WFD-ACC-44 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9838

Description

The Whitefield POP's Arista Access Node WFD-ACC-44 has experienced a critical optical transport failure, as evidenced by alarm code ALARM_9838. This failure domain is specific to the Access role and Arista vendor equipment due to the inherent design of the 400G Ethernet connectivity. The optical transport failure is likely caused by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The Arista EOS software, which operates on the device, utilizes the OAM (Operations, Administration, and Maintenance) protocol to monitor the optical transport links. However, in the event of a failure, the OAM protocol may not be able to detect the issue, leading to a delayed or incomplete alarm notification. As a result, the network administrator must take proactive measures to diagnose and resolve the optical transport failure to prevent network outages and data loss. This SOP outlines the steps to diagnose and resolve the optical transport failure on the Arista Access Node WFD-ACC-44.

Pre-requisite Checks

- Pre-requisite: Verify that the Whitefield POP is accessible and dispatch remote hands if necessary., Description: Ensure that the remote hands are on site and ready to assist with the troubleshooting and repair process.

- Pre-requisite: Check the redundant link status to determine if the failed link is part of a redundant pair., Description: Verify that the redundant link is functioning correctly to prevent network outages in the event of a link failure.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status:, Description: This command displays the status of all interfaces on the device, including the optical transport links. The output will indicate if the interface is up or down, and if it is part of a redundant pair.

Command: show interface optics, Expected Output: Optics Status:, Description: This command displays the optics status for each interface, including the SFP/XFP details and the received and
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transmitted power levels.

Command: show oam status, Expected Output: OAM Status:, Description:
This command displays the OAM status for each interface, including the
OAM protocol version and the last OAM event timestamp.
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Resolution Steps

Step 1: Step: Replace the failed SFP/XFP, Command: replace interface , Description: Replace the failed SFP/XFP with a new one, ensuring that it is compatible with the interface and the device's EOS software version.

Step 2: Step: Verify the fiber connection, Command: show interface optics, Description: Verify that the fiber connection is secure and that there are no signs of damage or degradation.

Step 3: Step: Adjust the Rx/Tx power levels, Command: set interface power-adjust, Description: Adjust the Rx/Tx power levels to ensure that they are within the optimal range for the interface and the device's EOS software version.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface Status: up,
Description: Verify that the interface is up and functioning correctly
after the resolution steps have been completed.

Command: show interface optics, Expected Output: Optics Status:
nominal, Description: Verify that the optics status for the interface
is nominal, indicating that the SFP/XFP is functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: replace interface original, Description: Restore the original SFP/XFP to the interface, ensuring that it is compatible with the interface and the device's EOS software version.

- Step: Re-adjust the Rx/Tx power levels, Command: set interface power-adjust, Description: Re-adjust the Rx/Tx power levels to their original settings, ensuring that they are within the optimal range for the interface and the device's EOS software version.